GIT and GITHUB

1. **Git and Github :** It is an open source - **source code management tool and it is a distributed version control system that tracks changes in source codes.** Git we should install in our local machine. It creates the local repo, from the local repo. We can push our codes to the remote repo which is the Github repository (kind of cloud storage).
2. **What is git workflow :** In our pc/server we should install the git then create a directory and get into the directory run the git init command once the commands run it creates the local repo and main branch. Then you can edit the files and add the files to the staging area using git add <filename> and then commit the changes using git commit -m command, it creates the commit Id based on your commits. Then, If you want to push the changes to the remote repo, add the remote repo url then use the git push command to push the changes to remote repo which is Github.
3. **Explain the git commands we use :** The git commands which I commonly use, Git add, Git commit , Git push, Git fetch and Git pull, Git Reset, Git Revert.
4. **explain pull and fetch : Git fetch** downloads changes from remote repo without merging But **Git pull** does the same action like fetch but merges into the current branch. This time conflict will occur, that's why we commonly used git fetch.

5. **Rebase and merge : Rebase** is like a cut and paste, If you use rebase command the copy is not available in the source, it's only available in the destination where we did the rebase.

   Merge : When we merge from one branch to the other branch. the copies available in both the source branch and the destination branch like copy and paste.

6. **Revert and Reset : Revert -** Whenever we do the revert operation the files last changes will be deleted and do the undo operation, **Reset -** In Reset we have a two operation which is soft reset and hard rest. When we do the soft reset then it performs the undo operation and the files last changes are also retrieved, if we do the hard reset both operations will perform undo from commit and the entire file will be deleted. So best practices we go with the soft reset.
7. **Explain your git branching strategy:** Yeah!. Branching strategies we follow in our organization which is like we have a different kind of branches which are **main branch, Feature branch, staging branch, release branch and hot-fix branch**. In the main branch, we have a successful code or a healthy code in the particular main branch. Then comes to the feature **branch**, all the developers work here for new features and for future updates, once they developed the code. It moves to the **Staging branch** and all the testing processes would be done here. Once the codes were tested successfully it is delivered to the **Release branch,** from the release branch the code is moved to the production server that is live environment. From the live environment the end user can access the application. If they're facing any issues in the live, they stop the current code and switch to the existing code which is already available in the

main branch. Due to this time the bug code will be troubleshooting by the developers on the separate branch which is the **hot fix branch.**

8. **how to resolve your git conflict :** Im not faced any conflict in my experience but my developers used that word continuously. I saw how they resolved that. Can I explain it? Once the conflict occurred they used a **merge tool** and compared the new code and existing code and fixed the conflicts manually.
9. **I have a set of commits and I want to copy a single commit:** To copy a single commit, use git cherry-pick <commit-hash>. It applies the commit changes to your current branch.
10. **What is git stash :** Git stash command is used to vanish the files in the staging area. It stores like a hidden file and whenever we run the stash-pop commands. It shows the vanished files again.
11. **I'm having a file with a set of 100 commits. Now I want to combine 50th to 100 commits :** To combine commits 50 to 100, use **git rebase** and **squash** commits 51–100 into commit 50. Alternatively, use git reset --soft followed by a new commit.
12. **How to set a username in git:** yeah! When installing the git you should do the configuration using the git config command. Like

    git config --global user.name "Your Name"

    git config --global user.email "your.email@example.com"

13. **How to delete branches in git:**Using the command like **git branch -d <branchname>** we can delete the branch using this command(local). **git push origin --delete <branchname>** (remote branch).

14. **How to create a PR request:** First we should create the branch and push the code. Then in github click the pull request, choose the base branch and compare with your branch, then add the title and description and hit submit, once you get approved from senior, click the merge pull request.
15. **what is git reflog command:** Git log which shows only the current history in commits. But reflog shows all the HEAD moves in the commit so we can retrieve the lost commits.
16. **how to copy a git repo from one account to another account:** To copy a repo, clone it from the old account, change the remote URL to the new account, and push all branches/tags. For a full migration, use git clone --mirror + git push --mirror.
17. **Git clone vs Git Fork:** Clone makes a local copy of a repo, while Fork makes a copy in your own GitHub account (for contributions or ownership).